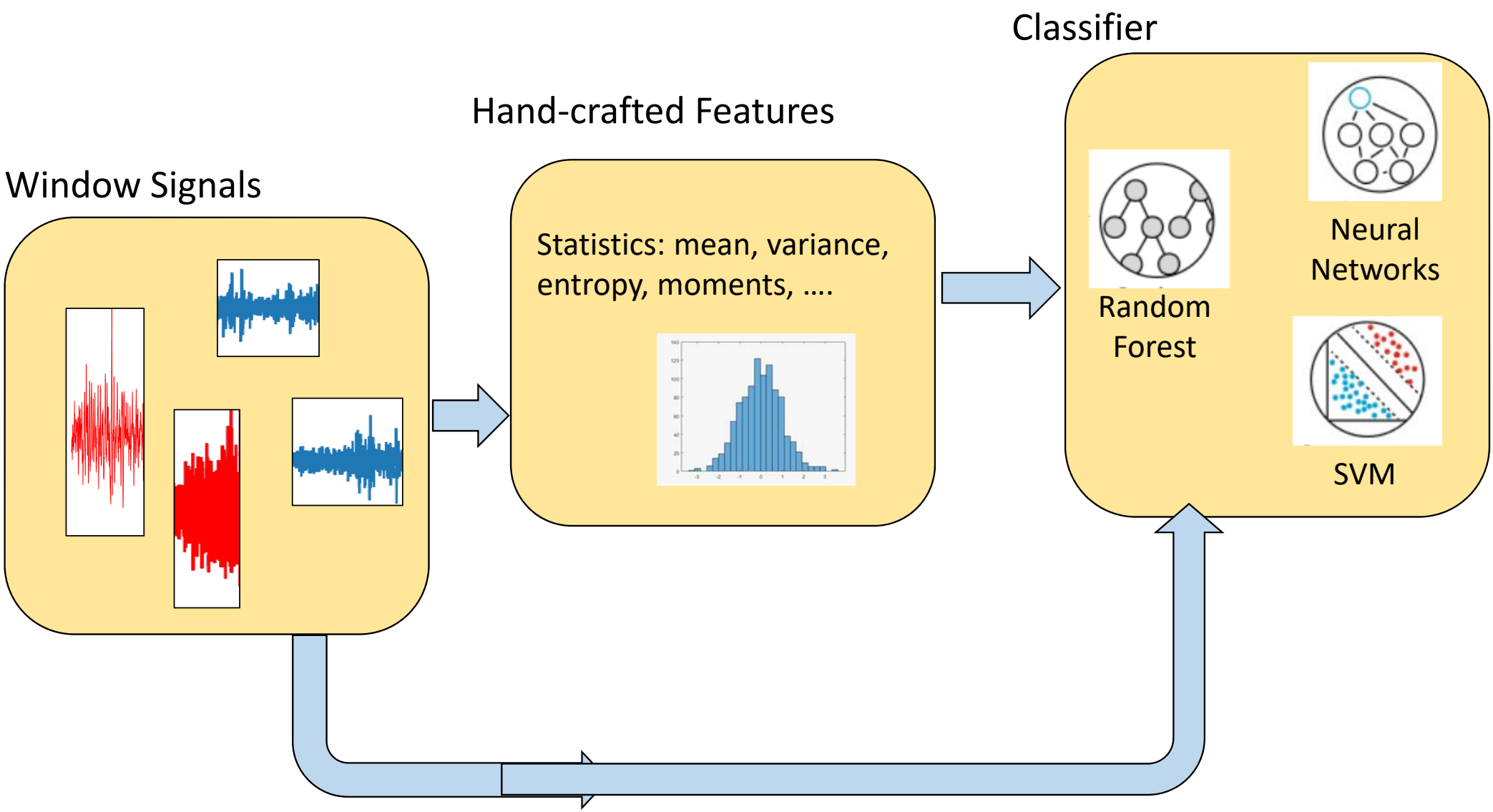
escola d'enginyeria

UAB

Classic Approach


Classifier
Hand-crafted Features
Window Signals
Statistics: mean, variance, entropy, moments, ...
Random Forest
Neural Networks
SVM

Hand Crafted Features

9 features describing signal pattern: mean, std, kurtosis, skewness, median, entropy, min, max, power_energy

File PatID_classicfeatures_seizure_EEGwindow_1.npz contains, for each subject an array of dimensions [NWindows, NChannels, NFeatures] =[NWindows , 21,9] such that
the i-th row [i,:,:] has the values of the 9 features computed for each EEG channel

These are the input features to a classifier

Hand Crafted Features

File PatID_seizure_metadata_1.parquet contains a dataframe with the following metadata associated to each window in PatID_classicfeatures_seizure_EEGwindow_1.npz:

1. *class*: window class (0,1)
2. *filename_interval*: number identifying, for each class, the interval the window belongs to inside recording (filename)
3. *global_interval*: number identifying, for each class, the interval in all recordings, the window belongs to
4. *filename*: filename identifying the recording

The variable *class* is the output target of the classifier.

Classifier

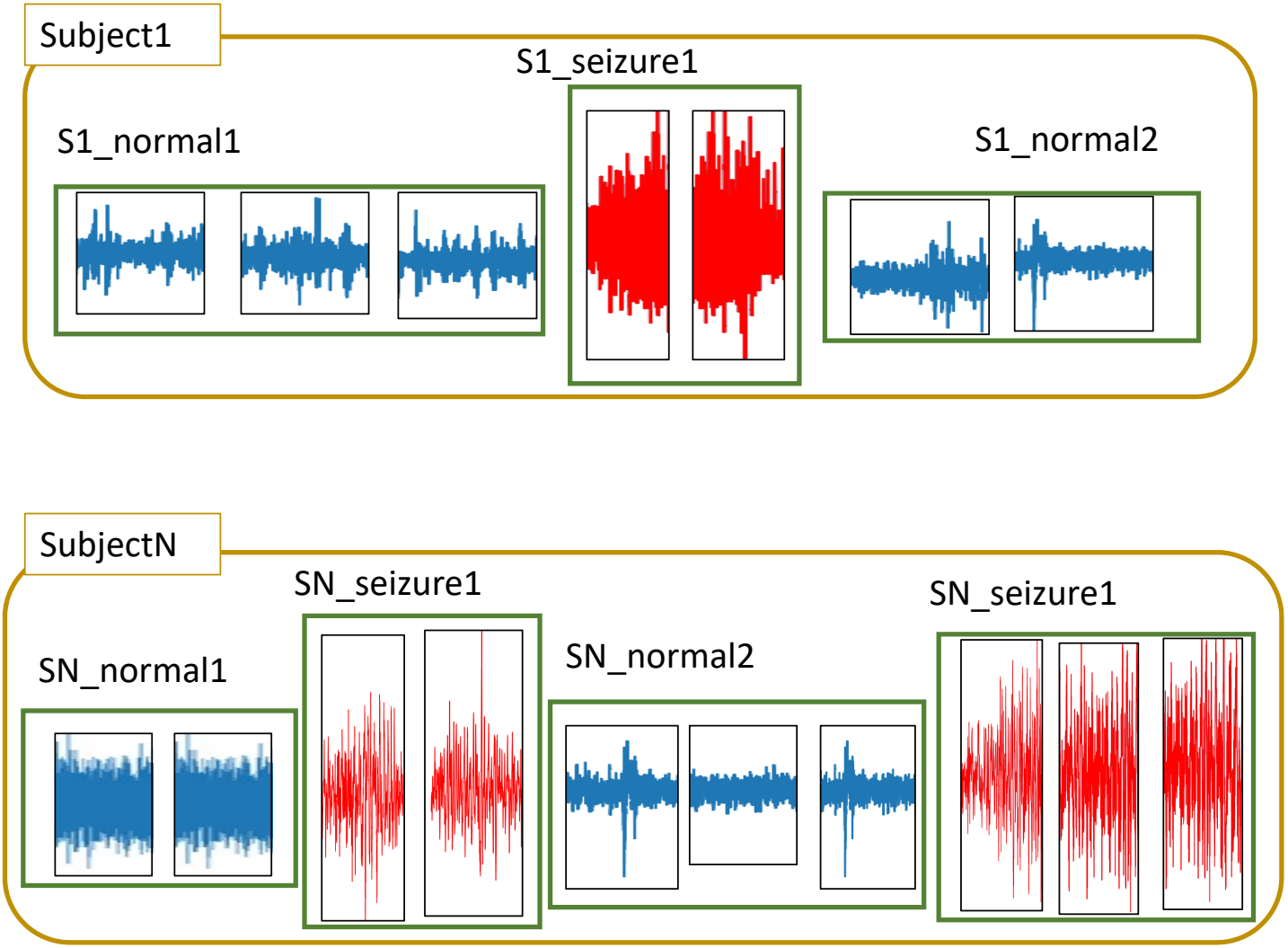
SVM with weighted loss:

```
from sklearn.svm import SVC

svm=SVC(class_weight='balanced',C=0.01,kernel='rbf')
```

SVM Hyperparameters: regularizing C, kernel function

Data Splitting

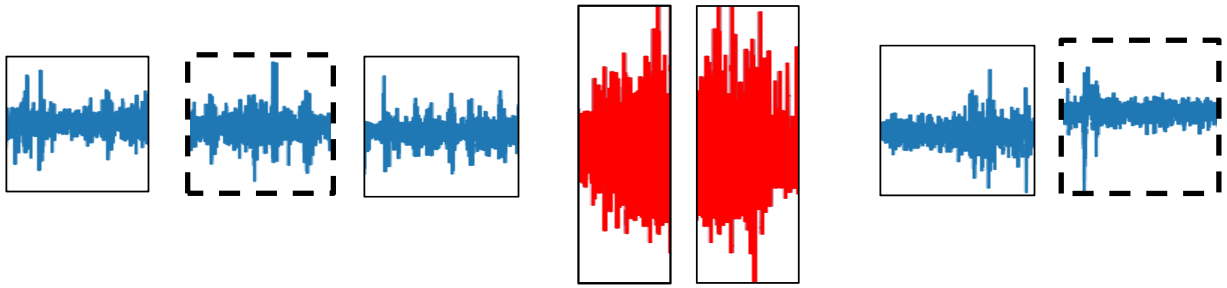


Sampling Levels:

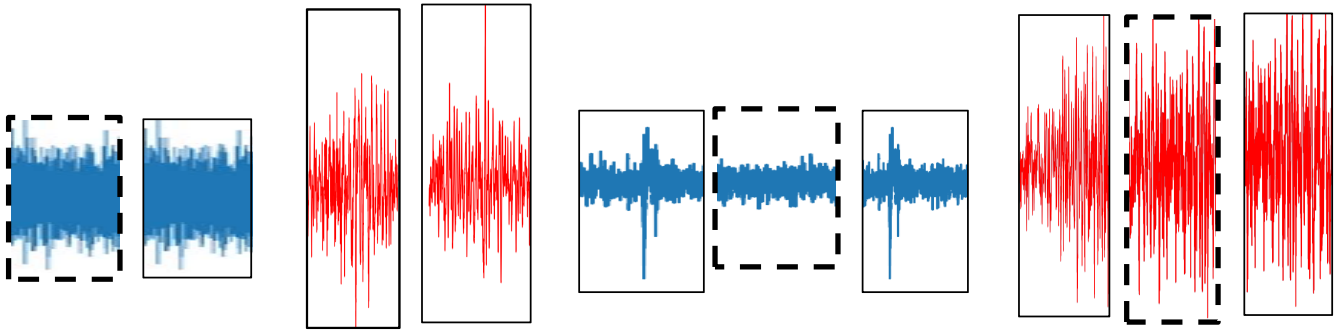
1. **Window**
2. **Seizure**
3. **Subject**

Data Splitting

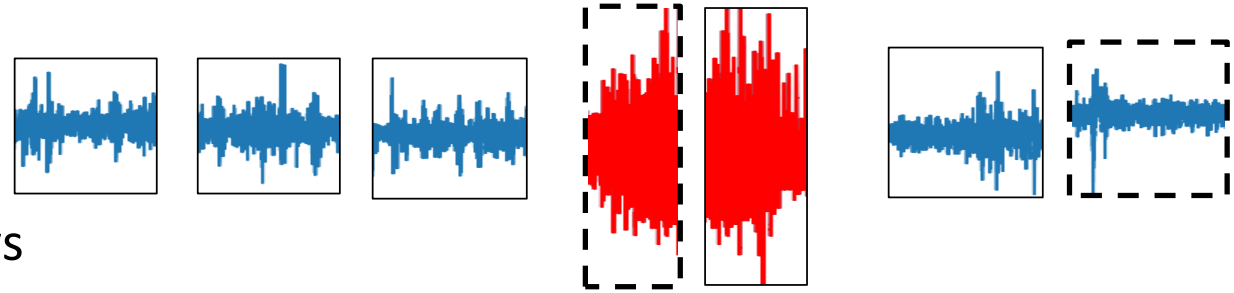
1. **Window Level:** K-fold on windows (experimental sampling unit)



- - test set



Data Splitting



1. **Window Level:** K-fold on windows

```
# Folds preserving proportion of classes in test set
from sklearn.model_selection import StratifiedKFold
skf = StratifiedKFold(n_splits=10)

# iteration over folds
for idxtr, idxtsin skf.split(np.arange(len(EEG_GT)), EEG_GT):
    # Train model
    svm=SVC(class_weight='balanced',C=0.01,kernel='rbf')
    X=EEG_Feat[idxtr[0][0::2],:]
    #X=X[:,ChannelSelection,:]
    X=np.reshape(X,(X.shape[0],X.shape[1]*X.shape[2]))
    y=EEG_GT[idxtr[0][0::2]]
     svm.fit(X,y)
```

Evaluation

62	**Train: summary undersampling Test balanced**							
63	**model**	**recall_0**	**recall_1**	**precision_0**	**precision_1**	**f1_0**	**f1_1**	**accuracy**
72	LSTM	0.9637+/-0.0015	0.9838+/-0.0005	0.9835+/-0.0005	0.9644+/-0.0014	0.9735+/-0.0009	0.974+/-0.0008	0.9737+/-0.0008
73	EpiGao19	0.9643+/-0.0029	0.9473+/-0.003	0.9482+/-0.0027	0.9637+/-0.0027	0.9562+/-0.0012	0.9554+/-0.0012	0.9558+/-0.0012
74	EpiHossain19	0.9338+/-0.0103	0.8604+/-0.012	0.8701+/-0.0087	0.9287+/-0.0091	0.9007+/-0.0014	0.8932+/-0.0026	0.8971+/-0.0017
75								
76	**Train: undersampling Test unbalanced (union res nores)**							
86	LSTM	0.9634+/-0.003	0.9838+/-0.0005	0.9996+/-0.0	0.369+/-0.0191	0.9812+/-0.0016	0.5365+/-0.0202	0.9638+/-0.0029
87	EpiGao19	0.9633+/-0.002	0.9473+/-0.003	0.9988+/-0.0001	0.3594+/-0.0116	0.9808+/-0.001	0.521+/-0.0121	0.963+/-0.0019
88	EpiHossain19	0.9318+/-0.0098	0.8604+/-0.012	0.9968+/-0.0002	0.2166+/-0.0197	0.9632+/-0.0052	0.3456+/-0.0248	0.9303+/-0.0094
89								

kfold cross validation | leave_one_patient_out | L1Seiz | L1Seiz separado | L1Seiz summ

Intervals computed for metrics obtained in test set